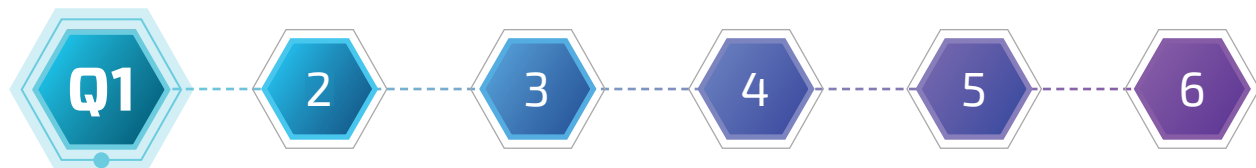


How To Tell If Your State Design is Accessible: The 6 Revealing Questions



What is the state in question?

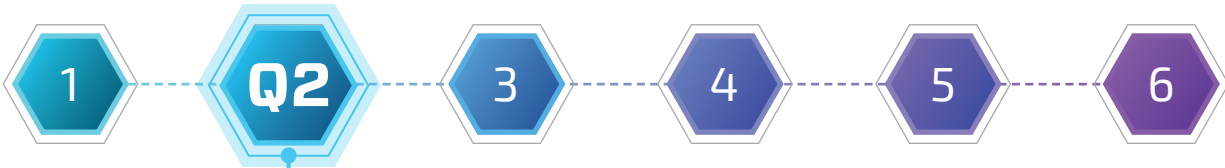
The hover state.

For reference

The 10 Fundamental States to Design:

1. Default / Enabled
2. Focus
3. Hover
4. Pressed / Active
5. Disabled
6. Selected / Toggled On (aka: activated, expanded)
7. Processing (aka: loading, working)
8. Abort (aka: cancel, terminate)
9. Visited — for links
10. Error — for form elements

*"Integrated
Activity Indicators"*



Is this an essential state?

(If no one could see this state, would it be detrimental?)

Yes/No and Why.

(If yes: continue with the questions.

If no: the remaining questions are N/A.)

Tip

Try answering
it both ways.

Terms

Essential

As defined by WCAG: "If removed, would fundamentally change the information or functionality of the content, **and** information and functionality cannot be achieved in another way that would conform." *Source: WCAG Glossary*

Detrimental

Primarily meaning: it would fail a WCAG accessibility inspection. Plus, secondarily meaning: it would fail known best practices in the accessibility community, whether they've been articulated in the latest version of the WCAG yet or not.

Tip

Since the answer to this question affects all of the following questions, try answering it both ways to help you think it through from both perspectives. Answer this question with "Yes, because..." and fill that in. Then answer it with "No, because..." and fill that in. Then see which answer feels like the strongest argument. Also, your answer could be a Yes in some aspects and a No in other aspects.

Is this an essential state?

Answer:

**Yes,
because...**

It improves usability for sighted mouse users. And for non-mouse interactions, a hover state is not expected.

The cursor turning to a hand is not a reliable indicator.

If we design it to be helpful for some, we need to make it helpful for all (no discrimination).

**No,
because...**

There is no specific WCAG guidance for designing hover states. If a website does not have hover states, it will still pass a WCAG audit. (No guidance, no failure.)

The compliant U.S. & U.K. design systems don't have color independent hover states.

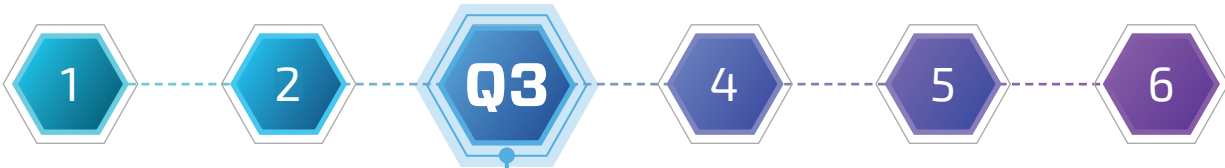
Google recommends making hover states subtle so they're not distracting (suggesting they're more decorative).

The default signifiers already indicate interactivity, thus reducing the need for hover states to confirm this.

Ultimate answer:

No (for WCAG compliance) & Yes (as a best practice).

No, it's not specifically called out in WCAG as being a required state. If you did not design this state, you would not fail an audit. However, it is an accessibility & usability best practice, since it does benefit sighted mouse users. And therefore strongly recommended to include where applicable.



Is the current design of this state color contrast compliant?

Yes/No and Why.

(If yes: continue with the questions.

If no: adjust the colors so they pass contrast requirements, and then continue with the questions.)

WCAG's criteria

For the informational part(s) of the element (such as a text label or a standalone icon)... They must maintain (and not lose) adequate color contrast with their backgrounds in their various states, excluding disabled. That's 4.5:1+ for text / standalone icons or 3:1+ for large-scale text (18 point or 14 point bold). *Source: WCAG Success Criterion 1.4.3: Contrast (Minimum)*

For the graphical part(s) of the element (such as a colored background, a border, a shadow, or whichever primary graphical styling indicates the state)... The visual information that is required to identify the states of a UI component (except inactive states), must have a contrast ratio of at least 3:1 against adjacent color(s).

Source: WCAG Success Criterion 1.4.11: Non-text Contrast

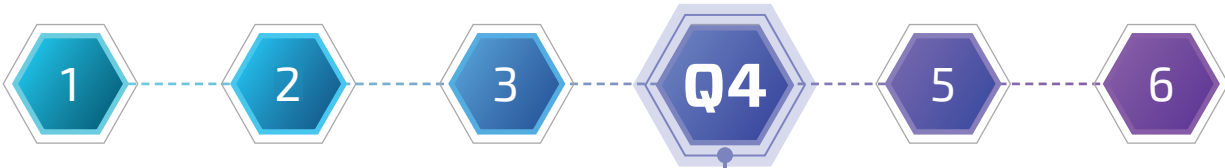
See for yourself

What would your interface design look like to someone with impaired vision (permanent, temporary, or situational)? Give yourself a general idea by using a browser Vision Simulator (such as the NoCoffee Vision Simulator) to look at your design. For this topic of color contrast, turn on pervasive issues such as Contrast Loss, Glare, and Cloudiness.

Tip

Remember to look at your state design on all the interactive elements where it will appear, not just on primary buttons.

[illegible]



Is the current design of this state color independent?

(Meaning, it does NOT rely on color alone to convey the state change or essential meaning.)

Yes/No and Why.

(If yes: continue with the questions.)

If no: make the needed changes to ensure it's color independent, and then continue with the questions.)

WCAG's criteria

To be color independent, a state needs a structural design change, such as an underline or border (as opposed to solely a color change). *Source: WCAG Success Criterion 1.4.1: Use of Color*

Alternatively, side-by-side relative luminance may be used (where the two states have a 3:1+ contrast ratio between each other, and the difference between the two can be seen side-by-side). *Source: WCAG Success Criterion 1.4.1: Use of Color > Sufficient Techniques > G183*

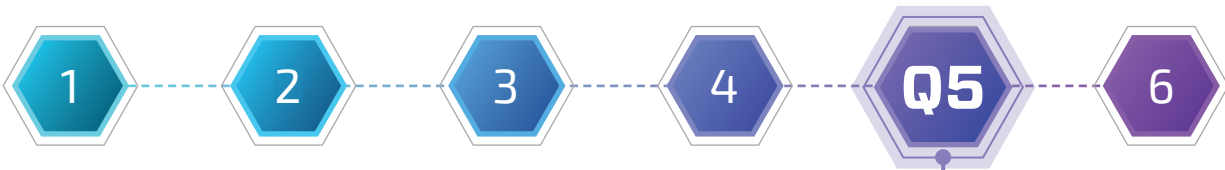
See for yourself

Use a browser Vision Simulator and turn on Achromatopsia (no color). Alternatively, turn your design grayscale using the design program of your choice.

Tip

When in doubt, test it out. Do usability testing with people who have complete color blindness to see if your design will work or not (this is particularly useful if you're attempting to use relative luminance to make a design color independent).

Type of interactive element	Is its state design Color Independent?	Notes (e.g. what to fix)



Is the state change sizable enough to be seen? (E.g. Is it obvious, noticeable, or large enough.)

Yes/No and Why.

(If yes: continue with the questions.)

If no: make the difference between the states more pronounced so it's easier for anyone to perceive [increase the size of the state indicator or highly-contrasting area], and then continue with the questions.)

WCAG's criteria

Meet/exceed one (or both) of the following:

The WCAG Minimum Area guidance — For an outline to be noticeable it should be 1+ CSS pixel thick around the perimeter of the interactive element; or for a shape, it should be the area of a 4+ CSS pixel thick line along the shortest side of a minimum bounding box. (Note: This criteria is specific to Focus states, but the overarching logic is clearly inferred — Essential elements need to be sizable enough so people can see them.)

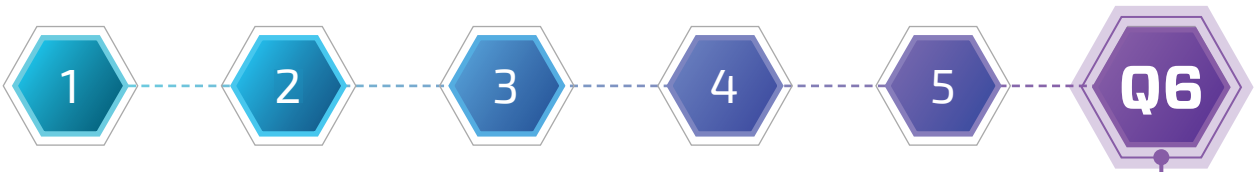
Source: WCAG Success Criterion 2.4.11: Focus Appearance (Minimum)

The WCAG Target Size guidance — For each target, there is an area with a width and height of at least 44 CSS pixels that includes it, and no other targets, except when the target is in a sentence or block of text. *Source: WCAG Success Criterion 2.5.8: Target Size (Minimum)*

See for yourself

Use a browser Vision Simulator, such as NoCoffee, and in the top box of pervasive issues, turn on Blur (low acuity). Try out some of the others as well.

Type of interactive element	Is its state change sizable enough?	Notes (e.g. what to fix)



Does this state use befitting signifiers to indicate its meaning?

(E.g. If a button state is supposed to be visually indicating that it's hovered, does it in fact look hovered? Usability testing will answer this.)

Yes/No and Why.

*(If yes: wonderful! Your design should be accessible now.
If no: make the needed changes until you can answer yes to this question.)*

Criteria

This question helps us ensure our design is appropriately styled and meets visual expectations based on the type of state (or element) it is. We want to make sure we don't design a state that could be mistaken for some other state.

Note: As of yet, there is no specific WCAG criteria that covers this best practice. However, Don Norman, co-founder of the Nielsen Norman Group (a source of authority on usability) provides a reputable and reliable source we can reference for this best practice. [See: *jnd.org/signifiers_not_affordances*](https://jnd.org/signifiers_not_affordances)

See for yourself

Search the internet for images of the state you're designing to see if yours is wildly different. Being different is not bad, it just means it could be unexpected for your users. Next, ask other people (outsiders that are not on your project team) if they can guess which state your design represents. If no one can guess it correctly, you may be looking at a future usability issue.

Type of interactive element	Does it use befitting signifiers to indicate the intended state?	Notes (e.g. what to fix)

Summary

Q1	Q2	Q3	Q4	Q5	Q6
What is the state in question?	Is this an essential state?	Is the state design color contrast compliant?	Is the state design color independent?	Is the state change sizable enough to be seen?	Does this state use befitting signifiers?
<i>E.g. Hover</i>	Yes/No	Yes/No	Yes/No	Yes/No	Yes/No

Remember

Aside from the best practices covered via The 6 Revealing Questions, the other things that make a state accessible (and usable) are the ones you already know that we talked about in the earlier sections, such as label wording, styling, hierarchy, etc.

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